CS4433 Moderated Learning Management System
Release v1.0

Release version: v1.0
Release type: Initial deployment release
Project: CS4433 Moderated Learning Management System
Programming language: C++17
Database: SQLite
Build system: CMake

1. Release Overview

The Moderated Learning Management System (LMS) v1.0 is an initial release developed to support the management of academic activities within an educational institution.

The system provides a role-based environment for Students, Lecturers, Heads of Department (HODs), and Administrators. It supports course enrolment, assignment management, electronic submissions, grading, moderation, and the release of student results.

2. Features

User Authentication and Account Management

User login and logout.

Role-based access to system functionality.

Password hashing and verification using libsodium.

Account creation and user management.

Student Functionality

- View available courses.
- Enroll in courses.
- View enrolled courses.
- View assignments.
- Submit assignments.
- View submission statuses.
- View released marks and feedback.

Lecturer Functionality

- View courses within their department.
- Create assignments with descriptions, deadlines, and maximum marks.
- View student submissions.
- Grade submissions.
- Provide feedback.
- Grant individual assignment extensions.

Head of Department Functionality

- View submissions awaiting moderation.
- Review lecturer-awarded marks.
- Adjust marks when necessary.
- Record reasons for mark adjustments
- Release moderated results to students.

Administrator Functionality

- Create Student, Lecturer, HOD, and Administrator accounts.
- View registered users.
- Manage academic departments.
- Add departments.
- Manage courses.
- Add courses and associate them with departments.

3. Assignment Submission and Late Penalties

The system supports electronic assignment submissions and applies a graduated late submission penalty policy.

Submission time	**Penalty**
On or before the deadline	0%
1 day late	10% of maximum mark

Submission time	Penalty
2 days late	20% of maximum mark
3 days late	30% of maximum mark
More than 3 days late	Final mark is 0

Lecturers can also grant individual extensions, with the revised deadline and reason recorded.

4. Moderated Grading Workflow

The system implements a moderated grading process:

- Students submit assignments.
- Lecturers review and grade submissions.
- Heads of Department review and moderate marks.
- Approved results are released.
- Students access their released marks and feedback.

The workflow uses the following submission statuses:

- Submitted
- Graded
- Moderated
- Released
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5. Technologies Used

- C++17
- SQLite
- CMake
- libsodium
- MSYS2 / MinGW

6. Quality Assurance

The project includes automated tests and a GitHub Actions continuous integration workflow.

Quality assurance activities include:

- Automated testing using the project's test suite.
- Static analysis using Cppcheck.
- Code metrics analysis using Lizard.
- Manual source-code review.
- Build verification through CMake.

The associated quality assurance reports are included in the project repository.

7. Deployment

The application is distributed as a C++ console-based application.

The installation guide provides instructions for obtaining the source code, installing dependencies, configuring the project, building the application, initialising the database, and creating the initial Administrator account.

Separate user manuals are provided for:

- Students
- Lecturers
- Heads of Department
- Administrators

8. Release Information

Version: v1.0
Release: CS4433 Moderated Learning Management System
Repository: https://github.com/Kevin-droid408/CS4433-Moderated-Learning-Management-System